# Certified SAT Covering-Radius Measurement: Representation Invariance and Lazy-Constraint Experiments

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**Abstract**

This technical report consolidates two follow-up studies to *Algorithmic Resolution and Geometric Distinguishability in SAT Solution Spaces* [8]. Both concern the *covering-radius resolution* $\hat{\varepsilon}(A, F)$ of a recorded sample $A$ of satisfying assignments: how far the worst solution of $F$ sits from the nearest recorded one. First, we record exactly how $\hat{\varepsilon}$ behaves when a formula is re-encoded with auxiliary variables. Measured on the original variables it is exactly invariant; measured on all variables it can move, by at most the number of auxiliary variables, and that bound is attained. The additive inequality is classical, appearing for codes in Cohen, Karpovsky, Mattson and Schatz [6]; what we add is the solution-relative form, with both projection hypotheses stated explicitly, and a margin criterion telling which certified comparisons survive a re-encoding. Applied to our own certified data, 43 of 116 comparisons carry no guarantee beyond zero auxiliary variables. Second, we report a small paired implementation study of two changes to the measurement procedure itself: tightening a lower bound to a witness's actual distance, and generating distance constraints lazily. The tightening is sound but recovers nothing on any of 257 reconstructed measurements. Lazy constraint generation is exact and needs only 3–43 % of the distance constraints. On the three units whose radius sequences we verified identical it reduces total time including certification from 46.0 s to 18.9 s; a fourth unit was exact under both methods in the original run but sits on the measurement budget and did not reproduce, so it is excluded from paired claims. On the five units that do not resolve, the lazy method is worse. We claim no general speedup, no asymptotic improvement, no solver ranking, and no complexity-class consequence.

## 1 Scope, and what is new

This is a consolidation report, not a new theory. It is written to be read alongside the antecedent paper [8], which introduced the observable, the experimental programme, and the solver comparisons that motivated everything here. Three things in it are new relative to that paper:

1. an explicit statement of the two hypotheses under which $\hat{\varepsilon}$ is preserved by re-encoding, and of the sharp additive bound when it is not (§3);
2. a margin criterion for certified *comparisons* of two samples, and its measured distribution over our own certified data (§4);
3. a paired implementation study of two changes to the measurement procedure (§6–§7).

Two things in it *qualify* earlier interpretations. The antecedent paper reported a solver ordering on $\hat{\varepsilon}$; a later budget-and-selection control found that ordering to depend on the selection rule as much as on the solver, and the cross-solver predictive claim of the earlier work has been retracted in the repository's claim ledger. Nothing in this report reinstates either.

Throughout we keep four categories visibly distinct: **theorems** (proved here or quoted with attribution), **finite validation** (exhaustive checks over small instances), **measured results**

(this study's runs), and **open questions**. Passing a finite test does not prove a universal claim, and we never present one as if it did.

*Remark* 1 (A naming caution). "Covering-radius resolution" is a *geometric observable*: a distance in the Hamming cube. It has nothing to do with *resolution* as a propositional proof system. We use the longer phrase, or simply $\hat{\varepsilon}$, wherever confusion is possible, and we never abbreviate it to "resolution" alone.

## 2 The observable and the exact query

Let $F$ be a CNF formula on $n$ variables with satisfying set $S_F = \mathrm{sat}(F) \neq \emptyset$, and let $A \subseteq S_F$ be a nonempty *recorded sample*: the distinct solutions some procedure actually wrote down. Write $d_{\mathrm{H}}$ for Hamming distance.

**Definition 2** (Covering-radius resolution)**.**

$$\varepsilon(A, S_F) \;=\; \max_{x \in S_F} \min_{a \in A} d_{\mathrm{H}}(x, a).$$

This is the smallest $\rho$ for which balls of radius $\rho$ around the recorded sample cover every solution. It measures *how well $A$ covers $S_F$ geometrically*, which is a different question from *what fraction of $S_F$ lies in $A$*: a sample can have tiny coverage and small $\varepsilon$, or large coverage and large $\varepsilon$. Coverage and covering-radius resolution are logically independent, as established in the antecedent work [8]. For a code $C \subseteq \{0,1\}^n$ and $S_F = \{0,1\}^n$ this is exactly the covering radius of coding theory [5].

**The exact threshold query.** $\varepsilon$ is computed, not estimated, through a family of SAT queries:

$$Q_r(x) \;\equiv\; F(x) \;\wedge\; \bigwedge_{a \in A} d_{\mathrm{H}}(x, a) \geq r + 1. \tag{1}$$

Each distance constraint is a cardinality constraint over all $n$ variables, encoded with the sequential counter of Sinz [11] via PySAT [10]. The query family is monotone in $r$, and

$$Q_r \text{ satisfiable} \Rightarrow \varepsilon \geq r + 1, \qquad Q_r \text{ unsatisfiable} \Rightarrow \varepsilon \leq r.$$

*Finite validation* 3 (Boundaries, exhaustively checked). Over all 3 679 pairs $(A, S_F)$ arising from $n = 4$: the ladder reconstructs $\varepsilon$ exactly and is monotone in $r$ in every case; $Q_0$ is unsatisfiable exactly when $A = S_F$, i.e. exactly when $\varepsilon = 0$; and $Q_n$ demands distance $\geq n + 1$ in an $n$-cube, so it is unsatisfiable unconditionally and yields the sound but vacuous $\varepsilon \leq n$. This is a finite check at one small $n$, not a proof for all $n$; the $r = 0$ and $r = n$ statements are immediate from the definitions.

**What makes a bound certified.** A *lower* bound requires a satisfying assignment that is independently re-checked against $F$ and against the *entire* recorded sample $A$. An *upper* bound requires a refutation: Glucose 3 [1] emits a clausal proof in DRUP form, which is checked by `drat-trim` [12] at a pinned commit. An `UNKNOWN` result, or an UNSAT whose proof was not checked, tightens *no* certified bound and is reported as such. Every interval in this report is bracketed by evidence of these two kinds.

## 3 Representation invariance

Let $G$ be a formula on $n + p$ variables and let $\pi : \{0,1\}^{n+p} \to \{0,1\}^n$ drop the auxiliary block. Two hypotheses are used throughout; *both* are required and neither implies the other.

**(H1) the solution set projects:** $\pi(S_G) = S_F$.
**(H2) the recorded sample projects:** $\pi(A') = A$ for the lifted sample $A' \subseteq S_G$.

(H1) is projected logical equivalence, which is exactly what projected knowledge compilation certifies: Bryant, Tan and Heule [2] partition variables into data and existentially quantified ones and certify, with HOL4-validated soundness, that a compiled representation is logically equivalent to the quantified input. Their *Skolem assignments construct* a lift satisfying (H2) for a sample one chooses — which covers every lift used below — but they do not certify (H2) for an *arbitrary recorded* sample. When $A'$ is whatever a solver happened to record, (H2) is an assumption about the recording procedure.

*Remark* 4 ((H2) is set equality, not injectivity). Failure of injectivity of $\pi$ on $A'$ does *not* violate (H2). Take $A' = \{(0,0),(0,1)\}$ and $A = \{0\}$: then $\pi(A') = \{0\} = A$, so (H2) holds while $\pi$ collapses two members. (H2) fails exactly when the projected set *omits or adds* an assignment relative to $A$. Non-injectivity is a separate matter: it changes distinct-sample counts and can change sampling weights, which affects cardinality-matched comparisons but not $\varepsilon$.

**Lemma 5** (Invariance on the original coordinates)**.** *Assume* (H1) *and* (H2)*. Then*

$$\varepsilon_x(A', G) \;:=\; \max_{v \in S_G} \min_{w \in A'} d_{\mathrm{H}}(\pi v, \pi w) \;=\; \varepsilon(A, S_F).$$

*Proof.* $d_{\mathrm{H}}(\pi v, \pi w)$ depends on $w$ only through $\pi w$, and $\pi$ maps $A'$ onto $A$ by (H2), so the inner minimum equals $\min_{a \in A} d_{\mathrm{H}}(\pi v, a)$. By (H1) the outer maximum ranges over exactly $S_F$. □

This is a restatement in metric language of the standard projection fact, and we claim no novelty for it. Its practical content is that a measurement taken on the original variables is untouched by auxiliary ones.

**Theorem 6** (Auxiliary-coordinate bound)**.** *Assume* (H1) *and* (H2)*, and measure on all* $n + p$ *coordinates. Then*

$$\varepsilon(A, S_F) \;\le\; \varepsilon_{\mathrm{full}}(A', G) \;\le\; \varepsilon(A, S_F) + p,$$

*and both bounds are attained.*

*Proof.* Since $d_{\mathrm{H}}(v, w) = d_{\mathrm{H}}(\pi v, \pi w) + d_{\mathrm{H}}(v|_{\mathrm{aux}}, w|_{\mathrm{aux}}) \ge d_{\mathrm{H}}(\pi v, \pi w)$, the lower bound follows from Lemma 5. For the upper bound fix $v \in S_G$, put $x = \pi v \in S_F$, choose $a^*$ attaining $\min_a d_{\mathrm{H}}(x, a) \le \varepsilon$, and take $w \in A'$ with $\pi w = a^*$, available by (H2); then $\min_{w'} d_{\mathrm{H}}(v, w') \le d_{\mathrm{H}}(x, a^*) + d_{\mathrm{H}}(v|_{\mathrm{aux}}, w|_{\mathrm{aux}}) \le \varepsilon + p$. For attainment take free padding, $S_G = S_F \times \{0,1\}^p$: lifting every $a \in A$ to every auxiliary value gives $\varepsilon_{\mathrm{full}} = \varepsilon$, and placing all lifts at one auxiliary value gives $\varepsilon + p$. □

**Attribution.** The additive-$p$ inequality is **classical**. Cohen, Karpovsky, Mattson and Schatz [6], Corollary 1, state that puncturing a code on $p$ coordinates reduces its covering radius by at most $p$ — the same phenomenon read in the opposite direction. There the maximum defining the covering radius runs over the *whole* Hamming space; Theorem 6 is the solution-relative form, where it runs over $S_F$ and the ambient set is coupled to the extension by (H1), and its proof is the classical one. Theorem 6 specialises to their statement when $S_F = \{0,1\}^n$. We emphasise that a limited literature search does not establish priority; we searched the covering-code and projected-counting literatures and report what we found, not an exhaustive survey.

*Finite validation* 7 (Sharpness and controls, exhaustively checked). Over small instances with exact integer arithmetic: variable permutations composed with coordinate complementations preserve $\varepsilon$, $\varepsilon_{\mathrm{full}}$ and $|A|$ exactly (153 600 (isometry, instance) pairs, no discrepancy); a uniquely determined duplicate variable $z = x_0$ adds no solution and leaves $\varepsilon_x$ untouched on all 120 tested pairs, yet moves $\varepsilon_{\mathrm{full}}$ on 80 of them, because the all-coordinate metric silently gives one original coordinate weight 2; free padding attains both ends of Theorem 6 (360 instances each); and

uniform sampling of $S_G$ projects to the multiplicity-weighted law, with exact total variation 0 for constant multiplicity and 1/6 for $\mathrm{mult}(x) = 1 + [x_0{=}1]$ at $n = 3$.

*Remark* 8 (What invariance does not say). Lemma 5 says a re-encoding cannot change the *value* of $\hat{\varepsilon}$ measured on the original variables. It does **not** say that a re-encoding cannot reduce the *cost* of computing that value. Preserving a quantity and cheapening its computation are compatible, and §7 measures a case where a changed representation is in fact cheaper. This report establishes no computational improvement from *projection* specifically; that is a gap in our evidence, not a theorem.

# 4 Comparison margins

Experiments compare two recorded samples on the same formula. Suppose $A_1, A_2 \subseteq S_F$ with a certified strict margin $M = \varepsilon(A_1, S_F) - \varepsilon(A_2, S_F) \geq 1$.

**Corollary 9.** *Measuring on all coordinates, under any extension by $p$ auxiliary variables with lifts of both samples satisfying* (H1) *and* (H2)*: if $M \geq p+1$ then $\varepsilon_{\mathrm{full}}(A_1') > \varepsilon_{\mathrm{full}}(A_2')$ for* every *such extension and lift; if $M \leq p$ then* some *extension and lift ties the comparison (at $M = p$) or strictly reverses it (at $M \leq p-1$).*

*Proof.* If $M \geq p+1$ then by Theorem 6, $\varepsilon_{\mathrm{full}}(A_1') \geq \varepsilon(A_1) = \varepsilon(A_2) + M > \varepsilon(A_2) + p \geq \varepsilon_{\mathrm{full}}(A_2')$. For the second part use free padding with $A_2$ lifted to one auxiliary value and $A_1$ to all, giving $\varepsilon_{\mathrm{full}}(A_2') = \varepsilon(A_2) + p$ and $\varepsilon_{\mathrm{full}}(A_1') = \varepsilon(A_2) + M$ exactly. □

**Corollary 10** (What a certified *interval* establishes)**.** *In practice $M$ is certified only as $M \in [\ell, u]$ with $\ell \geq 1$. Combining Corollary 9 with the interval gives three regimes:*

| *auxiliary variables* | *what the interval establishes* |
|---|---|
| $p < \ell$ | *$M \geq \ell > p$: every allowed extension preserves the ordering* |
| $\ell \leq p < u$ | *neither half applies: the interval alone does not settle it* |
| $p \geq u$ | *$M \leq u \leq p$: some allowed extension ties or reverses it* |

*An exact margin is the degenerate case $\ell = u$, where the middle regime is empty and the interval settles every $p$.*

*Remark* 11 (Two cautions). First, the second regime is an *existence* statement over the theorem's unrestricted permitted lifts. It does not say that a given encoding reverses a given comparison, and adding auxiliary variables does not automatically reverse anything. Second, the witnesses realising it use free padding with lifts of unequal cardinality. If one additionally *requires* matched lifted cardinalities or unique extensions, the sharpness claim is not established by the argument above and would need a separate proof; we do not claim it.

**Measured distribution.** Applying Corollary 10 to the certified comparisons recorded in this repository — 92 with an exact margin and 24 certified only as an interval, 116 in total — gives: 43 carry no guarantee beyond $p = 0$, of which 28 have an exact margin of 1 (so some single-auxiliary-variable extension ties them) and 15 are certified only as $M \geq 1$ with $u$ as large as 13 (so nothing follows at $p = 1$, but a void verdict does follow at $p \geq u$). These are *comparisons*, with budgets, selection schemes and replicates nested within at most 25 instances; they are not 116 independent observations. The practical recommendation is one sentence: *measure Hamming distance on the original variables, and report the margin beside the verdict.* The repository already does the former.

## 5 Encoding-dependent observables

The antecedent note on encoding-relative variance [9] considers a language LINSAT of satisfiable GF(2) linear systems and two polynomial-time encodings, and shows that a constraint-count observable has different gradient signs under them. Three scope points matter when citing it alongside $\hat{\varepsilon}$, and all three were checked directly:

1. **It varies an observable, not a decision.** LINSAT $\in P$ under both encodings. On 300 random systems the decision agreed between Gaussian elimination and brute force 300/300, while the observable differed on 295/300. Changing what a statistic reports is not changing what the problem decides.
2. **The closure encoding is exponential off a log-rank family.** The closure is spanned by whole equations, so $|\mathrm{Closure}(A, b)| = 2^{\mathrm{rank}[A|b]}$, the *augmented* rank. $2^{\mathrm{rank}\, A}$ is wrong in general: the inconsistent pair $x = 0$, $x = 1$ has $|\mathrm{Closure}| = 4$ while $2^{\mathrm{rank}\, A} = 2$. The two ranks coincide exactly on consistent systems (Rouché–Capelli). Since $\mathrm{rank}[A \mid b] \leq \mathrm{rank}\, A + 1$, the log-rank polynomiality conclusion is unaffected.
3. **GF(2) results do not transfer to CNF-SAT by renaming.** $|\mathrm{sat}(\mathrm{OR}_k)| = 2^k - 1 \neq 2^{k-1} = |\mathrm{sat}(\mathrm{XOR}_k)|$ for every $k \geq 2$.

We therefore cite [9] only for the point that *constraint-count* observables are encoding-relative. We do **not** treat it as an established barrier to resolving P versus NP, and we do not treat it as having predicted any subsequent result. Note also that $\hat{\varepsilon}$ lies outside its mechanism: $\hat{\varepsilon}$ is a function of the solution set, invariant by Lemma 5, and its encoding dependence (§3) comes from a different source — which coordinates the metric is taken over.

| source | witnesses | loose | max gain | mean gain |
|---|---|---|---|---|
| optimization (n=64 units) | 37 | 0 | 0 | 0.000 |
| budget/selection pilot | 557 | 79 | 6 | 0.316 |
| **total** | 594 | 79 | 6 | 0.296 |

Table 1: Stage 1: stored witnesses replayed against their recorded samples. A witness is *loose* when its actual distance exceeds the threshold that was requested.

| quantity | optimization | pilot |
|---|---|---|
| measurements reconstructed | 17 | 240 |
| with an improved certified lower bound | 0 | 0 |

Table 2: Stage 1: effect on the quantity the reported intervals use. No reconstructed measurement gains a better certified lower bound.

## 6 Study 1: witness tightening

Both measurement loops compute a witness's actual distance to the recorded sample and then update the lower bound only to the requested threshold $r + 1$. Since a witness $x \in S_F$ with $\min_{a \in A} d_{\mathrm{H}}(x, a) = d$ proves $\varepsilon \geq d$, the sound update is $\mathrm{lo} \leftarrow \max(\mathrm{lo}, d)$.

We replayed every stored witness against the recorded sample for its measurement, recovering samples through each stage's own hash-verifying loader (both the formula hash and the sample hash were required to match; nothing was repaired). Table 1 reports the result: of 594 witnesses, 79 are *loose*, with maximum gain 6; no witness produced a smaller bound, so the tightening is sound on every one. The hard $n = 64$ units contribute 0 loose witnesses out of 37.

| unit | $\|A\|$ | S total (s) | L total (s) | ratio | S width | L width | L constraints |
|---|---|---|---|---|---|---|---|
| n64_a2.5_k0_W | 150 | 119.3 | 80.6 | 1.48 | 0 | 0 | 18–44 |
| n64_a3.5_k0_W | 150 | 1.4 | 1.2 | 1.15 | 0 | 0 | 4–25 |
| n64_a4.2_k0_W | 144 | 2.6 | 1.7 | 1.49 | 0 | 0 | 15–30 |
| n64_a3.5_k1_W | 150 | 42.0 | 16.0 | 2.63 | 0 | 0 | 14–48 |
| n64_a2.5_k1_W | 150 | 134.6 | 120.2 | 1.12 | 11 | 11 | 23–64 |
| n64_a2.5_k1_eqW | 40 | 121.8 | 120.8 | 1.01 | 2 | 4 | 4–12 |
| n64_a2.5_k2_W | 150 | 121.2 | 121.2 | 1.00 | 11 | 15 | 62–62 |
| n64_a2.5_k3_W | 150 | 127.8 | 119.2 | 1.07 | 11 | 15 | 51–51 |
| n64_a2.5_k4_W | 150 | 119.5 | 120.7 | 0.99 | 11 | 15 | 62–62 |

Table 3: Stage 2, per unit. The first four rows are the units exact under both methods; the remaining five did not resolve. Ratios are computed from unrounded seconds.

| group | metric | S | L | ratio |
|---|---|---|---|---|
| verified paired (3) | total s | 46.0 | 18.9 | 2.43 |
| | solve s | 19.3 | 8.5 | 2.28 |
| | check s | 23.2 | 3.2 | 7.27 |
| | encode s | 1.4 | 4.0 | 0.34 |
| | queries | 10 | 10 | – |
| | summed width | 0 | 0 | – |
| exact under both (4) | total s | 165.3 | 99.5 | 1.66 |
| | solve s | 77.3 | 68.0 | 1.14 |
| | check s | 82.7 | 18.6 | 4.43 |
| | encode s | 2.1 | 7.1 | 0.29 |
| | queries | 14 | 14 | – |
| | summed width | 0 | 0 | – |
| unresolved (5) | total s | 624.9 | 602.1 | 1.04 |
| | solve s | 562.2 | 535.3 | 1.05 |
| | check s | 57.4 | 46.5 | 1.24 |
| | encode s | 2.5 | 11.6 | 0.22 |
| | queries | 17 | 10 | – |
| | summed width | 46 | 60 | – |
| all 9 units | total s | 790.2 | 701.6 | 1.13 |
| | solve s | 639.4 | 603.3 | 1.06 |
| | check s | 140.1 | 65.1 | 2.15 |
| | encode s | 4.6 | 18.7 | 0.24 |
| | queries | 31 | 24 | – |
| | summed width | 46 | 60 | – |

Table 4: Stage 2 paired totals. “Verified paired” are the units whose radius sequences and per-query outcomes were confirmed identical (Table 6). The all-units row mixes different amounts of completed work and is not offered as a headline.

| outcome | S | L |
|---|---|---|
| certified UNSAT upper bounds | 12 | 11 |
| verified SAT witnesses | 10 | 7 |
| unchecked UNSAT (moves no bound) | 1 | 1 |
| UNKNOWN | 8 | 5 |
| trivial UNSAT (arithmetic) | 0 | 0 |
| alarms | 0 | 0 |

Table 5: Stage 2 certification integrity across all 18 measurements.

| unit | hashes | S radii | L radii | identical | stop |
|---|---|---|---|---|---|
| n64_a2.5_k0_W | ok | 24,28,26 | 24,28,26,25 | NO | budget/exact |
| n64_a3.5_k0_W | ok | 12,14,15 | 12,14,15 | yes | exact/exact |
| n64_a3.5_k1_W | ok | 24,20,18,17 | 24,20,18,17 | yes | exact/exact |
| n64_a4.2_k0_W | ok | 12,10,9 | 12,10,9 | yes | exact/exact |

Table 6: Paired-per-query verification. One unit fails: its static ladder hit the budget, the sequences differ, and its exact answer did not reproduce.

Table 2 reports the quantity that matters. **No reconstructed measurement would have obtained a better certified lower bound.** The reason is the radius ladder: a loose witness at radius $r$ proves $\varepsilon \geq r + 1 + g$, but the search then issues a query at a higher radius that also succeeds, and the final bound is set by the highest successful radius.

Replaying each recorded radius sequence under the tightened rule, 21 of 557 pilot SAT queries and 0 of 37 hard-unit SAT queries would have been redundant. This is a *retrospective* count and a *lower bound*: after the first skip the policy would choose different radii, whose answers the replay cannot know. **It is not a measurement of online runtime saving, and we do not report one.**

**Verdict.** Adopt the tightening: it is a one-line change, sound on all 594 stored witnesses, and it cannot lose. Do not expect a speedup from it.

*Open question* 12. Every one of the 37 hard-unit witnesses has gain exactly 0. A plausible mechanism is that the cardinality constraint is binding — the solver satisfies $d_{\mathrm{H}} \geq r + 1$ tightly and has no pressure to exceed it — but we did not test this, and state it as an open question rather than an explanation.

## 7 Study 2: lazy distance-constraint generation

**The method.** Query (1) carries one distance constraint per recorded solution. Write $Q_r^B$ for the query carrying constraints only for $B \subseteq A$. The lazy method starts from a small $B$, solves $Q_r^B$, checks any candidate against the *whole* of $A$, and on failure adds the nearest violated $a \in A$ and re-solves. This is the classical constraint-generation loop of Dantzig, Fulkerson and Johnson [7], and the counterexample-guided abstraction refinement pattern of Clarke et al. [3, 4] in its SAT-based form [4].

**Exactness, stated before measuring.** $\mathrm{sat}(Q_r^A) \subseteq \mathrm{sat}(Q_r^B)$ for $B \subseteq A$, since $Q_r^B$ is a conjunction over a sub-family. Hence a *checked* refutation of $Q_r^B$ certifies $Q_r^A$ unsatisfiable, and a candidate satisfying $F$ with $\min_{a \in A} d_{\mathrm{H}}(x, a) \geq r + 1$ is a valid witness for $Q_r^A$ whatever $B$

produced it. Verification is always against the full $A$. Refinement adds the argmin, which the candidate violates, so no candidate recurs.

*Remark* 13 (What a lazy refutation establishes, exactly). When $Q_r^B$ is refuted and the proof checks, what has been established is: *the subset $B \subseteq A$ already suffices to exclude every satisfying assignment of $F$ at distance $> r$ from all of $B$* — equivalently, balls of radius $r$ about $B$ already cover $S_F$. It does **not** establish that $B$ is minimal with this property, that it suffices at any smaller radius, or that $\varepsilon(B, S_F) = \varepsilon(A, S_F)$. Each would need separate evidence, and we claim none of them. The identifiers of each $B$ and its relation to the full $A$ are archived with the run.

**Design.** Nine experimental units, each recovered with both hashes verified; two methods, static (S) and lazy (L), interleaved per unit with S first; identical radius policy, encoder, solver and checker; a 120 s per-measurement budget (reduced from the antecedent stage's 240 s, and stated in the protocol frozen before measurement), a 60 s per-invocation solver limit and a 120 s checker cap; initial $|B| = 4$ chosen by a deterministic greedy farthest-point pass; refinement capped at 60 iterations per query. Timing covers encoding, solving, proof checking and refinement overhead, and the budget is enforced on the measurement as a whole. Ratios are computed from unrounded seconds. All 18 measurements are reported; none was excluded.

**Results.** Tables 3–5 give per-unit figures, paired totals and certification integrity. The pattern is a dichotomy:

- **Four units reached an exact answer under both methods** in the original run, agreeing with each other and with the historical exact values. On these, total time including certification falls from 165.3 s to 99.5 s — though only three of the four survive the pairing check below. The gain is *mainly* certification-driven — checker time 82.7 s versus 18.6 s — though not only: solve time also falls slightly, and encoding time rises, since each refinement re-encodes.
- **Five units did not resolve under either method.** On four of them the lazy method ends with a *wider* certified interval, and on one it ties; summed final widths are 46 for static against 60 for lazy. The cause is visible in the iteration counts: the lazy loop spent 48–59 refinements inside a *single* radius query on three units, completing one query where the static method completed three.

Table 4 also reports the all-units aggregate, ratio $\approx 1.13$. **We do not offer this as a headline.** It mixes units where both methods finished with units where both merely exhausted the budget having completed different amounts of work; where neither finished, the meaningful comparison is interval width, and there the static method leads.

**Pairing, verified rather than assumed — and one unit fails it.** Equal query counts do not show that two methods asked the same questions, so we instrumented a re-run of the four units exact under both methods, recording the radius sequence and per-query outcome with the formula and sample hashes re-verified. Timings are *not* taken from that run; the reported seconds remain those of the original. Table 6 gives the result, and it is not uniform:

- **Three units are verified paired** — identical radius sequences, identical per-query outcomes, and the re-run reproduces the stage's certified interval, with both ladders terminating on their own rather than on the budget. On these three, total time falls from 46.0 s to 18.9 s (ratio 2.43), with checker time 23.2 s against 3.2 s.
- **One unit, `n64_a2.5_k0_W`, is not.** Its static ladder hit the measurement budget. In the original run it returned an exact answer $\varepsilon = 26$ at the budget boundary; in the re-run it stopped on the budget at $[25, 28]$ without reaching one, and the two methods asked different radius sequences ($[24, 28, 26]$ against $[24, 28, 26, 25]$). Its stage-2 "exact" outcome under the static method is therefore **not reproducible**, and the unit is excluded from every paired claim.

We report both groupings in Table 4 and let the reader see the difference. The four-unit figure of 165.3 s against 99.5 s is what was measured in the original run; the three-unit figure is what survives pairing verification. We do *not* promote the three-unit ratio as a better headline merely because it is larger: it rests on three units, and the excluded unit is excluded for a stated reason, not for its number.

**Constraints actually needed.** The lazy method terminated using 3–43 % of the recorded sample's distance constraints. *Fewer constraints do suffice.* The obstacle on hard units is the number of refinement iterations, not the final constraint count.

*Remark* 14 (A non-sequitur we had to retract). Removing constraints makes a refutation a logically *stronger* statement, since $\mathrm{sat}(Q_r^A) \subseteq \mathrm{sat}(Q_r^B)$. That is a statement about logical strength only: **it does not force proof length or runtime to increase.** On one unit the lazy proofs were indeed larger (524 081 versus 245 460 lines at $r = 31$, with a 3.3× smaller CNF), and we briefly inferred that this was the general direction. The study's own data refutes that: on the four resolved units the subset proofs are smaller and checker time falls by a factor exceeding four. That unit is a measured counter-case, not a law.

## 8 Limitations, and the remaining question

- This is a **small exploratory implementation study**: nine units at $n = 64$, one formula family, one solver, one encoding, one machine, no repetitions. It supports no population-wide claim, no asymptotic statement, no solver ranking, and no complexity-class consequence.
- The 120 s budget is half the antecedent stage's. Three of the four resolved units finished well inside it; the fourth, `n64_a2.5_k0_W`, sat on the budget boundary and did not reproduce. The five unresolved comparisons are budget-sensitive, and the widths quoted for them are specific to this budget.
- Study 1's query saving is retrospective, not an online runtime measurement.
- Theorem 6's sharpness is established for unrestricted permitted lifts; under matched lifted cardinalities or unique extensions it is open.
- Wall-clock time does not equalise algorithmic work across tools of different implementation maturity, and instances here were not independently selected.

*Open question* 15 (The remaining question). The two regimes are cleanly separated by one observable we already record: the refinement count. A method that begins lazily and reverts to the full encoding once refinements exceed a threshold would, on this data, keep the certification saving on the units that resolve and avoid the lost queries on the units that do not. Whether that holds beyond these nine units is untested, and we deliberately did not run it.

## Reproduction and evidence

All tables in this report are regenerated from the recorded data by `src/maketables.py`; none is typed by hand. The repository records the frozen protocol, the scripts, the per-measurement data, a claim-to-evidence map, environment and tool-version pins, and SHA-256 checksums for every artifact. Software versions actually used: CPython 3.14.7, python-sat 1.9.dev15, Glucose 3 via PySAT, and `drat-trim` at commit `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`.

A distinction worth stating plainly: the materials in the repository *regenerate the tables* from recorded measurements. Independently *re-verifying the numerical certificates* additionally requires the CNFs and clausal proofs, which are bulky; where those are not shipped in the Git tree, the report says so rather than implying full independent reproducibility.

## References


[1] Gilles Audemard and Laurent Simon. Predicting learnt clauses quality in modern SAT solvers. In *Proceedings of the 21st International Joint Conference on Artificial Intelligence (IJCAI)*, pages 399–404, 2009. Glucose. The solver used here is Glucose 3 through PySAT.

[2] Randal E. Bryant, Yong Kiam Tan, and Marijn J. H. Heule. Certifying projected knowledge compilation. In *28th International Conference on Theory and Applications of Satisfiability Testing (SAT 2025)*, volume 341 of *LIPIcs*, pages 8:1–8:22. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2025. DOI: `https://doi.org/10.4230/LIPIcs.SAT.2025.8`. Primary source read directly. Variables are partitioned into *data* variables and existentially quantified variables; the certifier establishes that the compiler output is logically equivalent to the quantified input formula, with soundness validated in HOL4 and checkers in C and CakeML. Skolem assignments represent how the quantified variables may be assigned given an assignment to the data variables. A full-text search of the paper for "Hamming", "distance", "covering radius", "diameter" and "metric" returns no occurrence.

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[7] G. Dantzig, R. Fulkerson, and S. Johnson. Solution of a large-scale traveling-salesman problem. *Journal of the Operations Research Society of America*, 2(4):393–410, 1954. DOI: `https://doi.org/10.1287/opre.2.4.393`. The constraint-generation pattern used here — solve a relaxation, check the solution against the full constraint family, add a violated constraint, repeat — is this classical row-generation loop.

[8] Elias De Jesus. Algorithmic resolution and geometric distinguishability in SAT solution spaces, September 2026. Version DOI `https://doi.org/10.5281/zenodo.23038050`; concept DOI 10.5281/zenodo.18216781. Verified against the Zenodo API on 2026-10-28: creator "De Jesus, Elias" (ORCID 0009-0007-0190-9143), published 2026-09-29, resource type "Technical note / publication", licence CC BY 4.0, single file algorithmic_resolution_sat_v2.pdf (646 145 bytes), which is byte-identical (sha256) to the copy read here. The record carries no version label and no related identifiers.

[9] Elias De Jesus. Encoding-relative variance (ERV) barrier, January 2026. Version DOI `https://doi.org/10.5281/zenodo.18269283`; concept DOI 10.5281/zenodo.18236865. Verified against the Zenodo API on 2026-10-28: creator "De Jesus, Elias" (ORCID 0009-0007-0190-9143), published 2026-01-16, resource type "Technical note / publication", licence CC BY 4.0, single file EncodingRelativeVarianceBarrier.pdf (367 386 bytes), byte-identical (sha256) to the copy read here.

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